Conservation repair — drift-correction as the actual fix

Status: computed and guarded — src/pacland/climate/drift_correction.py + tests/test_climate_drift_correction.py (9 tests, green). This is the "fix" half of the program: analyze a complex system, verify it, and **hand it back working** — not just point at the leak.

The finding, in one sentence

> **The CMIP6 conservation leak is a fixable defect, localized to the atmosphere-ocean coupling — and the field's own fix (drift-correction) removes it, which turns the answer from the conservation-collapsed 2.71 K (2 models) to the repaired 15-model 3.08 K.**

The repair

step	result
measure the leak	12/19 leak energy, 6/19 mass, 2/19 salt
localize it	energy/mass cross the air-sea boundary; salt (ocean) is conserved → the l
apply the fix	drift-correction — subtract the measured leak
re-verify	every model conserves by construction

python3 -m pacland.climate.drift_correction reproduces the block.

What "actually fixing" changes

Before the fix, the strict three-axis gate collapses to GISS-E2-1-G (2.71 K) — conservation is so strict that only one family passes. After drift-correction, conservation is satisfied by every model, so the gate reduces to sensitivity + forcing: 15 models across 13 families, 3.08 K. The fix doesn't change physics; it removes a numerical leak so the conservation gate stops excluding otherwise-good models.

The full fix, three tiers (not a half-measure)

tier	what	status
1 — global-mean	subtract the mean leak so the ledger closes on average	built
2 — zonal (spatial)	subtract the leak's latitudinal pattern so it conserves at every latitude	corrector + data from the mesh widening job (scripts/clim
3 — coupling patch	fix the non-conserving term in the model's own code	the builders' patch — leak_map() hands them the spa

The honest boundary, now smaller and sharper: tier 3 is the only part that is genuinely the model-builders' work, and even that we don't just hand off — we hand them the *leak map* (which latitudes leak, by how much) and re-check their fix.

Why this is the business

Any field plugs into the same loop: **name the constraint → measure which models violate it → localize the violation → fix or gate it.** The conservation repair is

the climate instance of the universal "analyze → verify → fix" offering.

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.drift_correction
PYTHONPATH=src python3 -m unittest tests.test_climate_drift_correction -v   # 9 tests
```

Data: data/climate/cmip6_drift.json (Irving et al. 2021). The trust board and ECS estimate are trust_board.py and ecs_model.py.